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# **FurShield: Master System Requirements & Enterprise Architecture Specification (SRS + PRD)**

**Project Name:** FurShield — Advanced Pet Care & Healthcare Ecosystem

**Document Version:** 2.0 (Final Enterprise Specification)

**System Architecture:** 4-Role Unified Pet-Centric RBAC

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## Table of Contents

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1. [Executive Summary & Ecosystem Philosophy](#) (#1-executive-summary--ecosystem-philosophy)
  2. [Core 4-Role Permission Matrix \(RBAC\)](#) (#2-core-4-role-permission-matrix-rbac)
  3. [Role 1: Pet Owner Portal \(Care, Health & Commerce\)](#) (#3-role-1-pet-owner-portal-specification)
  4. [Role 2: Veterinarian Portal \(Clinical Practice Management\)](#) (#4-role-2-veterinarian-portal-specification)
  5. [Role 3: Animal Shelter Portal \(Rescue & Adoption Lifecycle\)](#) (#5-role-3-animal-shelter-portal-specification)
  6. [Role 4: Administrator Portal \(Platform Governance & Analytics\)](#) (#6-role-4-administrator-portal-specification)
  7. [Cross-Cutting Shared Engines](#) (#7-cross-cutting-shared-engines)
  8. [Master Relational Database Architecture \(24+ Entities\)](#) (#8-master-relational-database-schema-24-entities)
  9. [End-to-End System Workflows & Interaction Sequences](#) (#9-system-workflow--interaction-sequences)
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## 1. Executive Summary & Ecosystem Philosophy

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**FurShield** is an interconnected, enterprise-grade **Pet Healthcare, Veterinary Management, Shelter & Adoption, and Care Planning Ecosystem.**

## ☀️ Central Design Principle: "Pet as the Master Graph"

System mein har module isolated nahi hai — har action **Pet Master Record (Pet ID)** ke ird-gird ghoomti hai:

- **Pet Profile → Medical History Timeline → Vaccine/Med Reminders → Care Score (0–100) → Vet Consultations & Rx → Digital QR Passport → Emergency Mode Card → AI Assistant Context.**

## 2. Core 4-Role Permission Matrix (RBAC)

| Feature / Module                       |  Pet Owner |  Veterinarian |  Animal Shelter |  Administrator |
|----------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Pet Registration & Digital Identity    | Full (Own Pets)                                                                             | Read-Only (Patients)                                                                           | Full (Shelter Pets)                                                                               | Full Admin Access                                                                                 |
| Medical Timeline & Document Vault      | Read / Upload                                                                               | Full Write / Edit                                                                              | Read / Edit (Shelter)                                                                             | Read / Audit                                                                                      |
| Vaccines & Medication Scheduler        | Log / Track                                                                                 | Prescribe / Update                                                                             | Log (Shelter Animals)                                                                             | Manage Standard Types                                                                             |
| Care Planner & Care Score (0-100)      | Full Management                                                                             | View in Consult                                                                                | Track Daily Routine                                                                               | Configure Weights                                                                                 |
| Digital QR Passport & Privacy Controls | Full Control                                                                                | Scan & Access                                                                                  | Generate for Rescue                                                                               | Monitor / Revoke                                                                                  |
| Emergency Mode 1-Click Trigger         | Activate / View                                                                             | Receive Emergencies                                                                            | N/A                                                                                               | Global Monitoring                                                                                 |
| Appointment Booking & Slots            | Book / Reschedule                                                                           | Manage Schedule/Slots                                                                          | N/A                                                                                               | Oversee All Bookings                                                                              |
| Clinical Consultation & Prescriptions  | View Prescriptions                                                                          | Full Clinical Entry                                                                            | View if Shelter Pet                                                                               | Audit / Review                                                                                    |
| Adoption Listings & Matching Engine    | Browse & Apply                                                                              | View Medical History                                                                           | Create & Manage Listings                                                                          | Moderate Content                                                                                  |
| Adoption Applications Workflow         | Submit & Track                                                                              | N/A                                                                                            | Review, Interview, Approve                                                                        | Dispute Resolution                                                                                |
| Product Marketplace &                  | Browse, Cart, Order                                                                         | N/A                                                                                            | Order Shelter                                                                                     | Full Catalog & Orders                                                                             |

| Feature / Module                   |  Pet Owner |  Veterinarian |  Animal Shelter |  Administrator |
|------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Cart/Orders                        |                                                                                             |                                                                                                | Supplies                                                                                          |                                                                                                   |
| AI Pet Care Assistant & Summarizer | Full Chat Access                                                                            | View Visit Summaries                                                                           | General Advice                                                                                    | Configure & Monitor                                                                               |
| Audit Logs & Platform Analytics    | N/A                                                                                         | N/A                                                                                            | N/A                                                                                               | Full Global Access                                                                                |

### 3. Role 1: Pet Owner Portal Specification

#### 3.1 Dashboard & Health Command Center

- **Live Health Cards:** Har pet ki photo, age, breed, weight, microchip ID aur **Dynamic Care Score (0–100)**.
- **Smart Alert Banners:** Automatic alerts (e.g., "*Rabies vaccine due in 4 days*", "*Give Amoxicillin dose at 8:00 PM*", "*Vet Visit with Dr. Ahmed tomorrow*").
- **Quick Access Fast Actions:** 🚨 *Emergency Mode*, ➕ *Log Care Task*, 📅 *Book Vet*, 🛂 *Show Pet Passport*.

#### 3.2 Digital Pet Passport & QR Code System

- **Pet Attributes:** Name, Species (*Dog, Cat, etc.*), Breed, Gender, DOB / Age, Weight, Blood Group, Microchip ID, Neutered status.
- **Privacy-Controlled QR Generator:**
  - System unique QR code generate karega.
  - **Configurable Visibility:** Owner decide karega ke scan karne par kya dikhe (*Public Contact, Allergies, Current Meds, Emergency Vet Details*). Confidential medical docs hide rahenge.
  - Shareable link with instant revocation capability.

#### 3.3 Longitudinal Health Timeline & Document Vault

- **Health Timeline:** Chronological visual history (*Birth → Vaccines → Vet Visits → Surgeries → Lab Tests → Weight Tracking*).

- **Medical Document Vault:** Categorized upload (*X-Rays, Blood Tests, Prescriptions, Vaccination Certificates*).
- **Document Attributes:** Document Type, Issue Date, Clinic Name, Vet Name, Notes, File Attachment.

### 3.4 Smart Vaccination & Medication Management Engine

- **Vaccination Scheduler:**
  - Vaccine Name, Dose Number, Date Given, Next Due Date.
  - Automated Reminders (30-day, 7-day, 1-day, and Due Day).
- **Medication Adherence Tracker:**
  - Medicine Name, Dosage, Frequency (e.g. 2x Daily), Start/End Dates, Instructions.
  - Logging: [Taken] , [Skipped] , [Missed] .

### 3.5 Intelligent Care Planner & Care Score (0–100)

- **Daily / Weekly / Monthly Task Planner:**
  - Daily (*Feeding, Walks*), Weekly (*Grooming*), Monthly (*Flea/Tick treatment, Vet visit*).
- **Care Score Engine (0–100):**
  - Algorithm calculates score based on up-to-date vaccines, recent checkup, task completion rate, and health profile completeness (with full transparency breakdown).

### 3.6 Emergency Pet Mode & Geolocation 🚨

- **1-Click Emergency Trigger:**
  - Instant red-alert card: Blood group, severe allergies, active medications, chronic conditions, microchip number, and emergency contacts.
  - Automatic listing of nearest 24/7 emergency veterinary clinics with direct call and directions.

### 3.7 Smart Vet Discovery & Appointment Booking

- **Vet Recommendation Engine:** Filter by location/distance, rating, specialization, and availability.
- **Booking Flow:** Pet Selection → Appointment Type (*General, Vaccine, Emergency, Surgery, Follow-up*) → Available Time Slot → Reason/Symptoms → Booking Confirmation.

### 3.8 Intelligent Adoption Marketplace & Application Tracker

- Browse shelter listings with high-resolution photos and temperament tags.
- **Compatibility Matching:** Fill preferences questionnaire → System calculates % Match Score.
- **Application Tracker:** Live status tracking (*Submitted* → *Under Review* → *Interview* → *Approved* → *Adopted*).

### 3.9 Pet Marketplace & Order Lifecycle (Scope-Compliant)

- Browse categories (*Food, Grooming, Healthcare, Toys, Accessories*).
- Add to Cart, update quantities, checkout order.
- Order Status Tracking: Placed → Processing → Ready → Completed / Cancelled.  
(*Payment gateway & live logistics excluded as per SRS*).

### 3.10 AI Pet Care Assistant & Vet-Visit Summarizer 🤖

- Context-aware conversational assistant knowing pet's breed, age, and allergies.
- Strict medical guardrails (*Educational advice with professional disclaimer*).
- **Vet-Visit Summary Generator:** Chat conversation ko structured doctor-ready notes mein convert karega.

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## 4. 🧑⚕️ Role 2: Veterinarian Portal Specification

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### 4.1 Vet Dashboard & Daily Patient Queue

- Today's patient list sorted by time slots with live status indicators ( Pending , Confirmed , In-Consultation , Completed ).
- Quick capacity stats (Total visits today, upcoming surgeries, emergency buffer slots).

### 4.2 Clinic Profile & Slot Availability Configuration

- Professional details: Qualifications, Specialization, Experience, Clinic Name, Address, Contact.
- Slot Management: Custom shift hours, slot duration (e.g. 20/30 mins), break buffers, and day-offs.

### 4.3 Comprehensive Appointment Workflow

- Appointment requests review: [Accept], [Reject with Reason], or [Reschedule with Proposed Slots].
- Complete Status Lifecycle: Booked → Confirmed → Checked-In → In-Consultation → Completed

### 4.4 Authorized Medical History & Clinical Examination

- **Patient History Access:** Appointment confirmed hone par pet ki past longitudinal history aur previous lab reports access karein.
- **Consultation Record Form:**
  - **Vitals:** Weight (kg), Temperature (°F), Heart Rate, Body Condition Score.
  - **Clinical Findings:** Symptoms, Physical Exam Observations, Diagnosis.
  - **Digital Prescription:** Drug Name, Dosage, Frequency, Duration (days), Special Instructions.
  - **Dietary / Care Instructions:** Lifestyle and nutrition plan.
  - **Follow-up Date:** Recommended next appointment date with automatic owner reminder.

### 4.5 Diagnostic Lab Requests & Review

- Lab Test Orders: Order tests (CBC, Blood Chemistry, X-Ray, Urinalysis) directly from consultation.
- Status tracking ( Requested → Sample Collected → In-Review → Completed ) and diagnostic remarks entry.

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## 5. 🏠 Role 3: Animal Shelter Portal Specification

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### 5.1 Shelter Dashboard & Live Capacity

- Total animals in shelter, available for adoption, under medical quarantine, pending adoption applications, total adopted.
- In-shelter daily tasks checklist (Feeding, grooming, medications for rescued animals).

## 5.2 Shelter Profile & Verification

- Organization name, registration license, contact person, phone, email, facility address, operating hours.

## 5.3 Rescued Animal Intake & Lifecycle State Machine

- **Animal Registration:** Intake source (*Rescued, Surrendered, Stray*), estimated age, species, breed, gender, temperament tags, multiple photos gallery.
- **State Machine:** Rescued / Intake → Medical Quarantine → Available for Adoption → Application Under Review → Interview → Approved → Adopted

## 5.4 Shelter Daily Care & Medical Records

- Feeding schedules, grooming logs, quarantine health notes, deworming, and shelter vaccination records.

## 5.5 Adoption Application Lifecycle Management

- Review applicant questionnaires (*Household size, living arrangement, existing pets, experience*).
  - Process actions: Shortlist → Schedule Interview → Request Reference → [Approve] / [Reject].
  - **Adoption Finalization:** Pet status update to Adopted, auto-generating Adoption Certificate and transferring pet profile to new owner.
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# 6. Role 4: Administrator Portal Specification

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## 6.1 Executive Analytics & Business Dashboard

- Platform metrics: Total Users by Role, Active Pets, Monthly Appointments, Adoption Success Rate, Marketplace Sales Volume, Low-Stock Alerts.
- Graphical analytical charts for platform health and engagement.

## 6.2 User Management & RBAC Governance

- Complete directory of Pet Owners, Vets, Shelters, and Admins.



- User actions: View Profile, [Activate], [Deactivate], [Suspend], [Delete], and Password Reset triggers.
- Review and approve newly registered Veterinarians and Shelters.

### 6.3 Content & Adoption Listing Moderation

- Monitor all public adoptable listings across shelters; remove fake or inappropriate listings.
- Moderate user-submitted reviews and ratings for Vets, Shelters, and Marketplace products.

### 6.4 Product Catalog & Order Management

- Manage product categories, add/edit/delete product items, prices, brands, SKUs, inventory counts, and image galleries.
- Order oversight: View all orders across the platform and update order processing statuses.

### 6.5 Care Knowledge Base Management

- Publish and edit pet care articles, species guidelines (*Dogs, Cats, Birds*), FAQs, and seasonal health tips.

### 6.6 Central Notification Broadcasting

- Broadcast platform-wide alerts (e.g. system maintenance, seasonal health warnings).
- Send role-targeted notifications (e.g. all Vets or all Shelters).

### 6.7 Enterprise Security & Audit Log System

- Immutable audit trail logging sensitive actions:  
Timestamp | User ID | Role | Action Type | Entity ID | IP Address | Status
- RBAC resource protection (Owner A cannot access Owner B's records; Vet can only access confirmed patients).

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## 7. Cross-Cutting Shared Engines

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### 7.1 Central Notification & Reminder Engine

- Event-driven triggers for Appointments, Vaccination due dates, Medication timers, Adoption application updates, and Order statuses.

### 7.2 Dynamic Care Health Score Engine (0–100)

- Formula-driven scoring algorithm with real-time explainability breakdown.

### 7.3 Privacy-Preserving Digital QR Passport

- Tokenized, owner-controlled, shareable emergency and identification profile.

### 7.4 1-Click Emergency Mode Engine

- Instant retrieval of critical medical facts, active medications, allergies, and nearest 24/7 clinics.

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## 8. Master Relational Database Schema (24 Entities)

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[ users ] 1—\* [ user\_roles ] —1 [ roles ] | | — 1— [ pets ] 1—1 [ pet\_passports ] | | |  
| — 1—\* [ health\_records ] | | — 1—\* [ vaccinations ] | | — 1—\* [ medications ] 1—\*  
[ medication\_logs ] | | — 1—\* [ medical\_documents ] | | — 1—\* [ care\_tasks ] 1—\* [  
care\_task\_logs ] | | — 1—\* [ care\_score\_history ] | | — 1—\* [ emergency\_events ] | | —  
1—1 [ veterinarians ] 1—\* [ vet\_availabilities ] | | | — 1—\* [ appointments ] 1—1 [  
clinical\_consultations ] | | — 1—\* [ prescriptions ] | | — 1—\* [ lab\_test\_orders ] | | —  
1—1 [ shelters ] 1—\* [ shelter\_animals ] 1—\* [ adoption\_applications ] | | — 1—\* [  
orders ] 1—\* [ order\_items ] —1 [ products ] —1 [ product\_categories ] | | — 1— [  
reviews\_ratings ] | | — 1— [ notifications ] | | — 1—\* [ audit\_logs ] | | — 1—\* [  
ai\_chat\_sessions ]

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## 9. System Workflow & Interaction Sequences

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### A. Clinical Consultation & Health Update Flow

1. **Owner** books appointment for Pet (*Max*) with **Dr. Ahmed (Vet)**.
2. **Vet** confirms slot; appointment status becomes `Confirmed`.
3. During visit, **Vet** opens consultation screen, enters vitals (*Temp: 101.5°F, Weight: 22kg*), diagnosis (*Mild Otitis*), prescribes ear drops (twice daily for 7 days), and sets follow-up in 10 days.
4. On consultation save:
  - Pet's **Health Timeline** auto-updates.
  - Ear drops auto-added to Owner's **Medication Scheduler**.
  - Follow-up date schedules an automated reminder for the owner.
  - **Care Health Score** recalculates.

### B. Shelter Rescue & Adoption Flow

1. **Shelter** intakes a rescued dog (*Buddy*), registers profile, health quarantine notes, and photos.
2. Once healthy, shelter updates status to `Available for Adoption`.
3. **Adopter (Pet Owner)** completes questionnaire; system shows *94% Compatibility Match*.
4. **Owner** submits adoption application.
5. **Shelter** reviews application, conducts interview, and marks `Approved`.
6. On adoption completion:
  - Pet's status transitions to `Adopted`.
  - Pet ownership automatically transfers to the Owner's dashboard.
  - New **Digital QR Passport** is instantly generated for the owner. '@ | Out-File -FilePath "FurShield\_Master\_Requirements\_Specification.md" -Encoding utf8